MILITARY SPECIFICATION
ASBESTOS FELT OR MAT, RESIN IMPREGNATED

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers types and classes of asbestos felt or mat impregnated with a resin at the semi-cured state.

1.2 Classification. The resin impregnated asbestos felt or mat shall be classified by type in accordance with 3.4.1 and by class in accordance with 3.4.2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-9299	Resin, Phenolic, Low Pressure Laminating
MIL-P-116	Preservation, Methods of

STANDARDS

Federal

Federal Test Method Standard 406	Plastics, Methods of Tests For

Military

MIL-STD-129	Marking for Shipment and Storage

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(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal, shall apply.

PUBLICATIONS

American Society for Testing and Materials

ASTM D 638-61T	Method of Test for Tensile Properties of Plastics
ASTM D 792-60T	Method of Test for Specific Gravity and Density of Plastics
ASTM D 1918-62T	Tests for Asbestos Content of Asbestos Textile Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the resin impregnated felt or mat shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials or processing of materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

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3.3 Material. The material shall be made by impregnating a Chrysotile asbestos fiber conforming to ASTM D 1918-62T, grade AAAA, with a resin conforming to MIL-R-9299, Type II, Class 2.

3 4 Physical and chemical properties

3.4.1 Types. The types of resin impregnated asbestos shall be as specified in Table I.

Table I - Types of Resin Impreganted Felt or Mat

	Nominal Thickness, Inches	Ounces per Square Foot
Type I	0.01	0.43 to 0.88
Type II	0.04	2.2 to 2.8
Type III	0.08	4.5 to 5.5
Type IV	0.15	9:0 to 11.0

NOTE: Separator not included in weights.

3.4.2 Classes. The classes of resin impregnated asbestos shall be as specified in Table II. All values are for individual specimens.

Table II. Classes of Resin Impregnated Felt or Mat

Class	Volatiles		Resin Content (percent)		Flow (percent)	
	Minimum	Maximum	Minimum	Maximum	Minimum	Maximum
1	9	19	44.0	60.0	28.0	38.0
2	8	18	37.0	49.0	24.0	33.0
3	7	15	37.0	49.0	17.5	26.5
4	7	15	32.0	42.0	11.0	19.0
5	6	14	28.0	38.0	11.0	19.0
6	3	11	22.0	32.0	2.0	7.0

3.4.3 Physical properties of cured material. The physical properties of the cured material shall be as specified in Table III. All values are for individual specimens.

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Table III. Physical Properties of Cured Material

Class	Specific Gravity, Minimum	Tensile Strength, Minimum, Psi	Compressive Strength, Minimum, Psi
1	1.15	13,000	6,000
2	1.30	21,000	12,000
3	1.50	26,000	13,000
4	1.55	28,000	14,000
5	1.70	32,000	14,000
6	1.80	35,000	14,500

3.5 Workmanship. The material shall be uniform in quality and shall be free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the asbestos felt or mat shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. A preproduction sample of sufficient quantity of the impregnated asbestos from each container manufactured in accordance with 3.1 shall be subjected to 3 each of the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the impregnated asbestos by the supplier, prior to the approval

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of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the tests as specified in 4.5.

4.4 Preproduction inspection. The preproduction sample, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations.

4.4.1 Laminating cycle and physical properties. Each class of material shall be subjected to the laminating cycle as specified in Table IV prior to evaluating the physical properties.

Table IV. Laminating Cycle

Class	Pressure, Psi	Time, Minutes	Temperature Degrees F
1	15 ± 2	30 ± 1	300 ± 5
2	50 ± 5	30 ± 1	300 ± 5
3	75 ± 5	30 ± 1	300 ± 5
4	100 ± 5	30 ± 1	300 ± 5
5	200 ± 10	30 ± 1	300 ± 5
6	500 ± 25	30 ± 1	300 ± 5

NOTE: All laminates shall have a post cure of 24 ± 1/2 hours at 300 ± 5 degrees Fahrenheit (F), 24 ± 1/2 hours at 350 ± 5 degrees F, and 24 ± 1/2 hours at 400 ± 5 degrees F.

4.4.1.1 Specific gravity test. The specific gravity of the cured material shall be determined in accordance with ASTM D 792-60T.

4.4.1.2 Tensile strength test. The tensile strength of the cured material shall be determined in accordance with ASTM D 638-61T.

4.4.1.3 Compressive strength test. The compressive strength of the cured material shall be determined in accordance with Federal Test Method Standard 406, method 1021.

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4.5 Quality conformance inspection

4.5.1 Visual examination. Visually examine each container in the lot to determine compliance with 3.5 and Section 5.

4.5.2 Volatile content test. Weigh a sample of approximately 5 grams (gm) to the nearest 0.001 gm. Dry to constant weight (approximately 15 minutes) at 325 ± 5 degrees F in an air circulated oven. Cool in a desiccator for 15 to 20 minutes, then accurately weigh sample to the nearest 0.001 gram. Calculations are as follows:

$$\text{Percent volatile} = \frac{\text{Weight loss}}{\text{Original weight}} \times 100$$

4.5.3 Resin content test. Use approximately two grams of sample which has volatile removed (see 4.5.2). Weigh sample to the nearest 0.001 gram. Place sample directly into furnace for 60 to 65 minutes at 1525 ± 25 degrees F. Remove sample from furnace and cool in a desiccator for 15 minutes, then weigh (to the nearest 0.001 gram) to constant weight. Calculations are as follows:

$$\frac{\text{Ash weight} \times 100}{\text{Original weight} \times 0.86} = \text{Percent inorganic}$$

Percent inorganic subtracted from 100 percent = percent resin content

(0.86 in above formula is a correction factor for water of crystallization of Chrysotile asbestos.)

4.5.4 Resin flow test. Cut sample material to be tested in nominal 4-inch by 4-inch squares. Stack up neatly as many plies as required to obtain weight of approximately 25 grams, weighed to the nearest 0.01 gram. Place material between waxed chrome plates (stainless steel plates may be substituted) and press at 300 ± 5 degrees F at 400 ± 25 psi until resin is completely cured. This would be approximately 5 minutes for class 6 materials and up to 30 minutes for class 1 materials. Remove all resin flash from pressed sample and weigh the sample to the nearest 0.01 gram. Calculation of flow is as follows:

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$$100 - \frac{(\text{pressed weight} \times 100)}{(\text{original weight})} = \text{Percent flow}$$

Preimpregnated felts or mats shall be conditioned before test at a room temperature of 70 ± 5 degrees F, and a relative humidity of 50 ± 10 percent for 12 hours prior to testing. If this is not possible, allowances must be made in readings. Temperature and humidity should be recorded at time of test if above conditions cannot be met.

4.6 Acceptance criteria

4.6.1 Preproduction. Failure of any sample to conform to any requirement of this specification shall be cause for rejection of the sample.

4.6.2 Quality conformance. Failure of any sample to meet any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified by the procuring activity, the preservation, packaging and packing shall be in accordance with MIL-P-116, method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as insulating material in rocket motors.

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6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Whether preproduction sample is required
d. Place of delivery
e. Size of container
f. Request for test data
g. Whether broad goods sample is needed for testing

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-A-81264(WP) Asbestos Felt or Mat, Resin Impregnated

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD, FORM 1 APR 63 1426

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